

Lyrical Landscapes: A Computational Exploration of Theme and Sentiment in Music (2000–2015)

POP77142: Quantitative Text Analysis

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ABSTRACT

This study applies computational text analysis to explore the emotional and thematic dimensions of popular music lyrics from the Billboard Hot 100 between 2000 and 2015. Using a hybrid approach that combines Latent Dirichlet Allocation (LDA) for topic modelling and VADER for sentiment analysis, the research identifies recurring lyrical themes and traces shifts in emotional tone over time. Results reveal that while conventional themes dominate the lyrical landscape, emotional nuance is often more complex than surface-level labels suggest. By splitting the dataset into positive and negative subsets, sentiment-specific topic modelling allowed for a deeper understanding of how tone influences thematic expression, highlighting patterns such as self-assurance in positive lyrics and emotional turmoil in negative ones. The study also identifies temporal sentiment trends linked to cultural and societal events, revealing a dynamic relationship between music and its broader context. While computational methods proved effective in identifying lexical patterns, limitations emerged in interpreting metaphorical and context-dependent language. The study concludes with suggestions for future research, including genre segmentation and supervised learning approaches, to better capture the richness of musical expression and enhance the interpretive power of computational tools in cultural analysis.

INTRODUCTION

Music has historically served as a powerful tool of artistic expression, offering a window into the cultural, emotional, and social climates of its time. Lyrics act as a form of storytelling, performing as creative endeavours that reflect and respond to ever-changing sentiments. The text, themes, and tones of lyrics provide data-rich materials for computational cultural analysis to gain valuable insights. This study aims to understand the thematic content and emotional tone of popular music at the beginning of the 21st century. This research focuses on songs from the Billboard Hot 100 between 2000 and 2015, a period marked by dynamic and diverse influences on the music industry, media, and society. Computational methods such as Topic Modelling and Sentimental Analysis can be used to uncover underlying patterns in lyrical content, revealing potential themes like love, identity, or power expressed over time. More specifically, Latent Dirichlet Allocation (LDA) helps identify recurring themes across song lyrics, while tools like VADER allow insight into the emotional tones embedded in texts. By integrating these two methodologies into a hybrid approach, Sentiment-Specific Topic Modelling, this study explores the interaction between emotional nuance and thematic content. Furthermore, this research aims to assess the effectiveness of the specified computational tools in analysing emotive, textual works, demonstrating their potential to offer data-driven perspectives and analytics. This study contributes to the academic field through its creation of a large-scale, longitudinal dataset of popular songs of the early 2000s. Its integration of topic modelling and sentiment analysis By combining both sentiment and thematic analysis, this study seeks to offer a deeper understanding of how emotional and thematic content manifested in popular music from 2000 to 2015, and whether these patterns can provide meaningful insights into the use of textual analysis for broader emotional and thematic landscapes of cultural texts.

LITERATURE REVIEW

Cultural studies often view music as an indicator of social moods and lyrics as cultural artefacts, with previous research applying lyrical analysis to explore topics such as politics, gender, and economic shifts, showcasing the role of music in life and the use of textual analysis to reflect these patterns. Pettijohn & Sacco’s (2009) lyrical analysis across societal and economic changes over time supports this idea that popular music acts as a cultural barometer. Brand et al. (2019) further establish the treatment of lyrics as reflective of broader cultural and emotional patterns in their textual analysis to trace emotional expression from the 1970s to the 2010s. Additionally, the use of natural processing languages (NLPs) and emotional sentiment analysis is justified through prior academia. Mahedero et al. (2009) carry out textual analysis via NLPs and thematic categorisations to build a framework for organising lyrical content and identifying topics, validating the feasibility and use of automated analysis to study lyrical structure. Nandwani and Verma’s (2021) review of sentiment analysis in music shows the applicability of VADER and other sentiment tools for cultural texts, however, it also highlights the limitations and challenges of capturing linguistic emotion and musical intonation from basic text. Together, these studies establish a strong foundation for the integration of computational tools in lyrical analysis, paving the way for more ad-

vanced techniques to uncover deeper thematic patterns in music. Topic Modelling is a crucial tool for the Social Sciences, with LDA providing an adaptable, consistent machine learning method to allow for the grouping of textual concepts (Ramage et al., 2009). This popular tool is used across disciplines due to its flexibility in analysing non-traditional text types, such as musical pieces (Pandur et al., 2020). By discovering latent themes and patterns in large-scale data, big-picture lyrical themes can be uncovered. Calcina & Novak’s (2022) study grouping artists based on shared lyrical themes demonstrates the usefulness of LDA and music. Similarly, Sentiment Analysis tools, such as VADER, LIWC, and Sentistrength, allow for clustering of data through emotional proximity, reinforcing the idea that emotion is complex and multidimensional (Choi et al., 2018). Schab & Kruspe’s (2024) analysis of lyrics includes audio analysis to create a holistic emotional profile, showcasing the notion that musical content is multi-layered in itself and requires further exploration to truly capture the essence of lyrical content. The combination of the two analytical methods, as seen by Riberio & Ramos (2023) on Brazilian music, shows how joint modelling can reveal more complex interpretive insights. Overall, there is a lack of studies that carry out this hybrid approach of looking at topics or theme breakdowns within sentiment categories, especially in the realm of music. This gap highlights the potential for integrated approaches that combine computational tools to unravel the nuanced layers of music lyrics, which this study attempts to understand and possibly fill.

METHODOLOGY

Data Collection The dataset was built using a Billboard Hot 100 song list curated by the University of Texas at Austin journalism program, dating from 1958 to the present day. This list was filtered to the target timeframe, with duplicated chart entries removed, and only title, artist, and year retained, resulting in 5,980 unique songs. Genre and lyrics were added using two APIs: Spotify and Genius. The Spotify API retrieved genre tags; however, these were ultimately not included in the analysis due to missing values and labels based on artists instead of individual songs. Lyrics were collected from the Genius API, subject to availability on the platform. A custom function was developed to loop through the Billboard dataset and match lyrics and genres to each song using the title and artists as keys. This collection process took over 15 hours, using `time.sleep()` to prevent API request rejections due to high volumes. After retrieval, non-lyrical content (contributor tags, metadata, special characters) was removed, and the cleaned dataset was exported as a CSV. The complete collection and cleaning process can be found in the Github Repository linked in the Appendix.

Text Processing Prior to analysis, the final CSV file underwent a series of preprocessing steps. Lyrics were converted to lowercase, and all punctuation and numbers were removed. Metadata markers (“Chorus” and “Verse”) were stripped, along with common musical fillers (“oh,” “ayy,” “la,” “na,” etc.) and general profanity, using regex-based filters. This step was crucial as such fillers were disproportionately influencing early topic modelling results. Standard stopwords were removed using NLTK, along with a custom list tailored to exclude genre-specific and non-semantic words. Lemmatisation was then applied to reduce words to their base forms. Bigrams (two-word phrases) were initially tested but discarded, as

they introduced excessive noise and reduced model clarity. These preprocessing decisions were made iteratively during the initial tuning phase and were applied consistently across all computational tasks. The code for this processing and the following textual analysis methods can be found in the Appendix under ‘Analysis code.’

Topic Modelling Topic Modelling was conducted in two stages—vectorisation and Latent Dirichlet Allocation (LDA)—using the *sklearn* library. *CountVectorizer* was chosen as it aligns better with LDA’s probabilistic framework, using raw word counts rather than normalised term weights like TF-IDF. The following parameters were used for vectorisation:

max_df = 0.95 to ignore extremely common terms

min_df = 5 to remove rare words and reduce noise

max_features = 5000 to manage dimensionality

Following vectorisation, LDA was run with varying topic counts (*n_components* = 8, 10, and 12) to evaluate coherence and interpretability. After empirical tuning, 12 topics were selected as the optimal balance, as 8 topics resulted in overly broad groupings while 12 offered more nuanced yet coherent themes without over-fragmentation. Each topic was interpreted by examining its top 10 contributing words and three representative songs. Themes were then manually labelled based on lexical patterns and lyrical context. Each song was assigned a dominant topic, based on the highest topic probability, which was recorded as a new feature for further analysis.

Sentiment Analysis Sentiment analysis was performed using VADER (Valence Aware Dictionary for sEntiment Reasoning), accessed through the *nltk.sentiment* module. VADER was selected for its efficiency and ability to handle informal, expressive text, making it well-suited to the stylised language of song lyrics. Lyrics frequently include slang, informal grammar, repetition, and unique phrasing—elements that VADER is specifically designed to interpret more accurately than traditional sentiment lexicons. It utilises a pre-trained lexicon with rule-based heuristics that account for capitalisation, intensifiers, and negations, all of which are important for capturing the tone and emotional expression in lyrics. The sentiment process involved applying the VADER analyser to the cleaned dataset. For each lyric, four sentiment scores were extracted: pos: proportion of positive sentiment neu: proportion of neutral sentiment neg: proportion of negative sentiment compound: a normalised overall sentiment score ranging from -1 to +1 Lyrics were then classified into sentiment categories (positive, neutral, negative) based on their compound scores. These sentiment labels were then added to the dataset, creating a framework for analysing overall emotional trends across songs.

Sentiment-Specific Topic Modelling To explore how emotional tone influences lyrical themes, a sentiment-specific topic modelling approach was used. Based on earlier sentiment analysis, the dataset was split into two subsets: Positive and Negative lyrics. Neutral sentiment was excluded due to its ambiguity. Each subset was vectorised using the same *CountVectorizer* configuration as in the general topic modelling process. LDA was then

applied to each sentiment group ($n_components = 6$), yielding six topics per subset, which provided a balance between interpretability and granularity. Topic distributions were extracted for each lyric, with the top 10 highest-weighted words were retrieved. Themes were once again manually labelled by identifying patterns across high-weight terms. This process produced two sentiment-specific theme dictionaries, revealing how emotional tone shapes the thematic landscape of popular music lyrics.

RESULTS

Topic Modelling Table 1 presents the results from the Latent Dirichlet Allocation (LDA) process, identifying 12 distinct themes across the dataset. Each topic reflects a unique lyrical thread, offering insights into common patterns. While the model was generally effective in grouping songs based on thematic content, there were some instances where songs did not fully align with the assigned theme. This suggests a limitation of LDA when applied to music: the model tends to group songs based on surface-level word association rather than deeper emotional intent or lyrical meaning.

A recurring thematic element across topics was love or heartache, spanning multiple emotional valences. This complexity underscores the multidimensional nature of song lyrics. Notably, themes related to dreams and idealism exhibited a high degree of coherence, suggesting that certain topics were more effectively captured by others. However, it is essential to note that in nearly every topic, there were songs that seemed out of place, underscoring the challenges of applying topical modelling to nuanced and poetic language. Songs often contain multiple thematic layers, with lyrics encompassing a range of emotions and concepts. For example, a song centred on love might also explore power dynamics, longing, or physical intimacy. Due to the poetic, non-linear, and metaphorical nature of song lyrics, relying on word counts alone is inherently limiting. The broader context—such as the genre, artist, and performance delivery—plays a crucial role in shaping lyrical meaning that is often lost in purely textual analysis. Therefore, it is not unlikely that certain songs are not correctly given in context by LDA. However, this does not reflect a failure of the model, but rather a reflection of the fluidity and complexity in musical language.

Table 1: Compact Summary of LDA-Derived Themes

Topic	Theme	Top Words	Example Song
1	Party & Club Vibes	stop, rock, party, music, keep, dem, said, ima, sing, goin	<i>On The Ocean</i> –K’Jon
2	Love, Trust & Emotional Tension	could, name, call, lie, tell, money, think, even, gotta, made	<i>Opposite Of Adults</i> –Chiddy Bang
3	Reflective / Searching for Meaning	still, high, live, good, god, well, matter, side, old, road	<i>Latch</i> –Disclosure ft. Sam Smith
4	Romantic Encounters	tonight, give, bit, wait, freak, babe, try, yes, left, gotta	<i>Happy</i> –Ashanti
5	Power, Ego & Provocative	tell, bad, money, move, really, lookin, put, okay, look, show	<i>Just A Lil Bit</i> –50 Cent
6	Heartache & Vulnerability	shes, tell, think, would, said, ever, could, leave, much, really	<i>Diary</i> –Alicia Keys
7	Missing Someone	gone, long, think, better, good, said, miss, around, without, home	<i>Go On Girl</i> –Ne-Yo
8	Energy & Escapism	put, run, hot, everybody, beat, ready, crazy, nothin, light, hit	<i>Easy</i> –Paula DeAnda ft. Bow Wow
9	Dreams & Idealism	day, world, every, could, find, eye, dream, light, look, fall	<i>The Only Exception</i> –Glee Cast
10	Sensual & Intimate Moments	body, gotta, kiss, touch, hold, tonight, breathe, head, gimme, slow	<i>Mr. Brightside</i> –The Killers
11	Movement & Confidence	shake, break, alright, work, hard, que, bounce, pop, booty, show	<i>Roll</i> –Flo Rida
12	Streetwise Attitude & Bragging	big, yall, low, drop, throw, black, club, real, hit, white	<i>Stay Fly</i> –Three 6 Mafia

Sentiment Analysis Figure 1 illustrates the overall distribution of sentiment across the dataset. Positive sentiment dominates, with approximately 58% of songs being classified in this group. Negative sentiment follows as the second most prevalent, comprising around 37% of the songs, while neutral sentiment is relatively rare, accounting for only about 5% of the songs. This distribution suggests that popular music tends to lean towards positive emotional tones, aligning with the general themes of uplift, love, and hope seen in the music industry. However, the notable presence of negative sentiment reflects the popularity of themes such as heartbreak, struggle, and introspection in contemporary music. The distribution of sentiment across the dataset confirms the idea that popular music is overwhelmingly positive but also nuanced with negative sentiment that speaks to deeper themes and inner turmoil. The findings suggest a dynamic relationship between music’s emotional tone and cultural contexts that shape it.

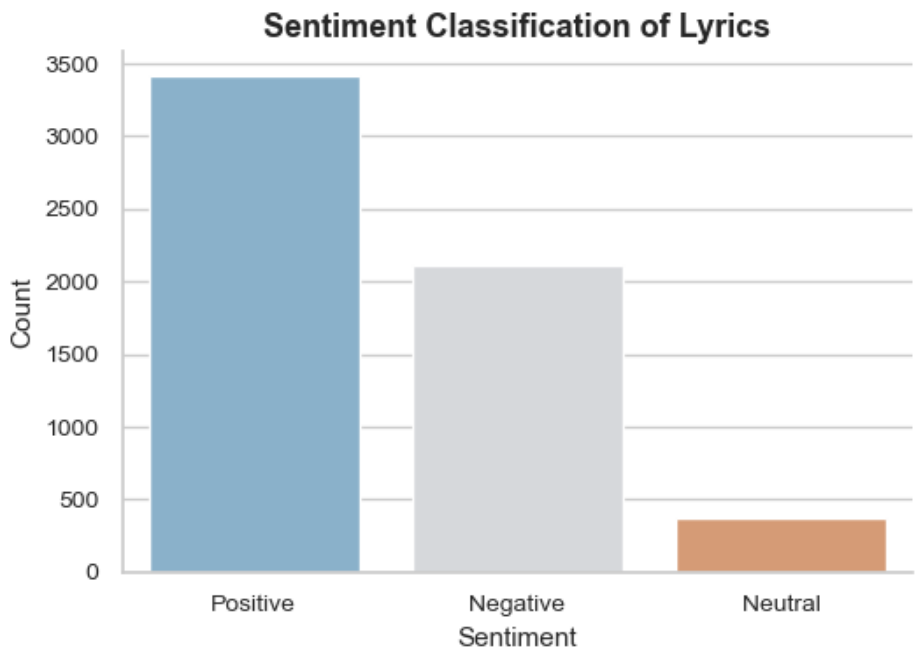


Figure 1: General Sentiment Analysis

Sentiment-Specific Topic Modelling The sentiment-specific topic modelling findings offer a deeper understanding of how emotional tone shapes the thematic structures within popular music lyrics. By separating the topics into positive and negative sentiment, as seen in Table 2, music’s complex and dynamic emotional landscape is revealed. The results show that positive sentiment is not confined to simplistic portrayals of love or happiness, and, similarly, negative sentiment extends beyond sadness or heartbreak. These findings challenge conventional expectations of emotional expression and highlight the highly contextual nature in which sentiments manifest in lyrics.

Table 2: LDA Sentiment Topics: Themes, Prevalence, and Top Words

Positive Sentiment Topics			
Topic	Theme	% of Songs	Top Words
Topic 1	Swagger & Hype Energy	6.8%	hand, tonight, okay, body, beat, music, turn, alright, bring, air
Topic 2	Optimism & Light	16.5%	kiss, light, day, hair, sun, sweet, world, long, fly, star
Topic 3	Love & Waiting	9.6%	girl, work, rock, hot, talk, gotta, ima, play, run, really
Topic 4	Fun, Fame & Social Buzz	8.2%	good, ready, tell, look, roll, think, ride, yes, body, better
Topic 5	Sensuality & Confidence	47.2%	day, think, tell, said, better, thing, friend, world, gotta, home
Topic 6	Friendship, Reassurance & Beauty	11.8%	money, party, big, girl, yall, hit, real, club, white, black
Negative Sentiment Topics			
Topic	Theme	% of Songs	Top Words
Topic 1	Stress, Confusion & Burnout	5.3%	shake, low, drop, wit, shes, bring, dirty, gotta, trouble, left
Topic 2	Hustle & Confrontation	18.4%	think, crazy, really, tell, lie, said, thing, stop, somebody, leave
Topic 3	Street Talk & Bravado	15.7%	money, big, hot, club, hoe, hit, yall, real, beat, break
Topic 4	Club Vibes & Seduction	38.7%	gone, world, eye, hold, try, day, long, pain, wrong, walk
Topic 5	Heartache & Disconnection	12.9%	tell, yall, head, ima, die, nothin, fuckin, bout, black, ride
Topic 6	Nostalgia & Longing	8.9%	bad, stop, day, high, light, hard, turn, lookin, good, home

For positive sentiment, the topics revealed a significant emphasis on themes of empowerment, self-assurance, and emotional reliance. Nearly half of all positive songs reflected themes centred around confidence, sensuality, and optimism, marking a shift in the portrayal of positivity. This evolution indicates that contemporary music has evolved beyond romantic or idealistic expressions to embrace a multifaceted understanding of positivity. Topics such as “Optimism & Light” and “Friendship & Beauty” suggest that positivity is not just an individual pursuit but also one rooted in social and aesthetic values. This shift is evident in the transition from themes of “waiting for love” to “owning one’s worth,” signifying a cultural move toward self-empowerment and confidence in contemporary music. On the other hand, negative sentiment revealed a broader emotional complexity than just sadness or heartbreak.

While these themes of loss and heartache remain prevalent, new patterns emerged tied to disconnection, detachment, and turmoil. Notably, topics such as “Club Vibes & Seduction” indicate growing trends of negative sentiment framed through experiences of nightlife, seduction, and posturing. These topics often result in individuals projecting outward confidence while grappling with inner issues. This trend is further reflected in the themes of “Hustle & Confrontation” and “Street Talk,” which suggest a characterisation of external aggression and bravado, possibly masking underlying vulnerability or conflict. These results illustrate that negative sentiment is far more nuanced, encompassing not only themes of loss and heartache but also tension, self-preservation, and emotional defence in a modern world.

Temporal Trends Compound sentiment was analysed over time (Figure 2) to explore potential shifts in lyrical content. The average compound sentiment fluctuated across years but remained positive overall, as the sentiment score never dipped below zero. A distinct peak in sentiment was observed in 2007, with the compound score reaching around 0.30, indicating a period of heightened emotional positivity. Notable dips in sentiment were recorded in 2004, 2008, and 2012, with the sharpest decline observed in 2015. This suggests that the emotional tone of music evolves in response to broader social, cultural, and industry trends. For example, the 2004 drop reflects the rise of emo subculture, the 2008 dip corresponds with the economic downturn and surge in themes of personal hardship and dissatisfaction, while the 2015 decline correlates with the growth of trap music’s aggressive tone and the rise in turmoil-driven, introspective pop.

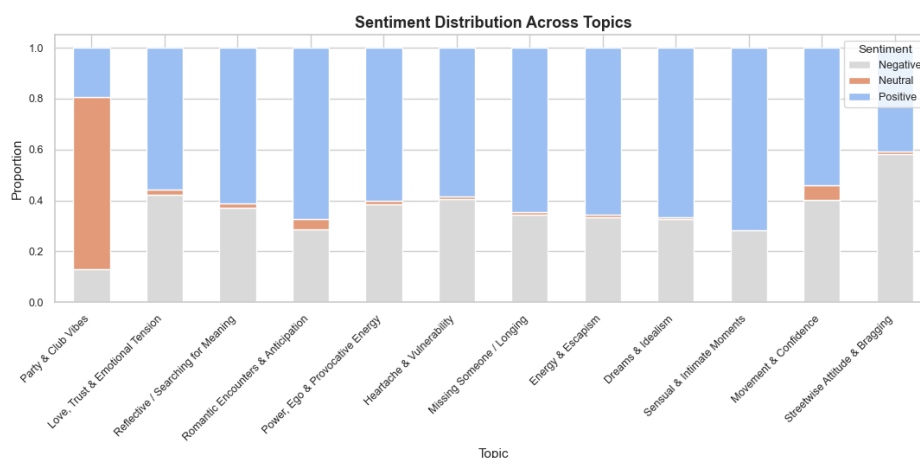


Figure 2: Compound sentiment scores by year

The positive sentiment trends, shown in Figure 3, reveal interesting shifts over time. “Friendship, Reassurance & Beauty” consistently emerged as the dominant theme from 2000 to 2015, with notable peaks in the early 2000s, as well as 2007 and 2014. This underscores the enduring appeal of emotionally affirming, supportive themes that resonate with listeners seeking connection, social validation, and personal upliftment in music. Following this, “Optimism & Light” saw a steady increase after 2010, culminating in a significant peak in 2015. This trend indicates a shift toward more hopeful and motivational themes, suggesting that post-2010, there was a cultural push towards messages of positivity. This increase in

resilience in popular music may be in response to global and societal challenges of the time. Meanwhile, themes like “Sensuality & Confidence” and “Love & Waiting” display greater fluctuation patterns, although both saw a noticeable increase in popularity after 2012. This suggests a growing embrace of self-expression and emotional vulnerability within positive music.

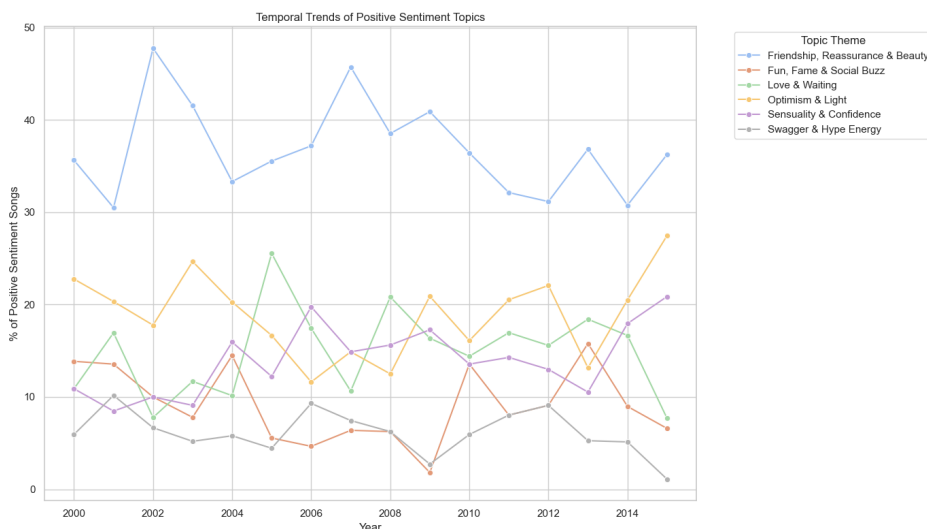


Figure 3: Positive Sentiment Topics Over Time

For negative sentiment trends over time (Figure 4), “Nostalgia & Longing” remained a dominant theme throughout the 2000s and early 2010s, with only minor dips. This reflects the lasting appeal of themes centred on reflection, loss, and melancholy in popular music. However, the rise of “Hustle & Confrontation,” particularly in latter years, signals a shift toward more socially and culturally charged narratives of struggle, ambition, and tension. The theme of “Heartache & Disconnection” shows volatility with notable peaks in 2008 and 2010, but a general decline in recent years. This suggests that themes of romantic loss have lost some ground to more complex emotional states. Finally, “Stress, Confusion & Burnout” rose sharply post-2010, aligning with growing awareness of mental health issues, though it still represents a small proportion of negative sentiment overall.



Figure 4: Negative Sentiment Topics Over Time

DISCUSSION

Strengths This study demonstrates how computational methods can reveal latent patterns in musical language, offering a fresh perspective on popular music’s emotional and thematic landscape. Topic Modelling proved effective in identifying recurring lexical themes across a broad range of songs. Topics such as club anthems, emotional vulnerability, and self-growth emerged with strong coherence in many cases. This indicates that, even with nonlinear text, certain vocabulary patterns remain consistent in genre-typical and emotion-laden lyrics. The preprocessing stage played a critical role in this outcome, as removing common fillers further clarified thematic distinctions, allowing for meaningful clusters to form. Furthermore, certain topics, “Romantic Encounters” and Streetwise Attitude,” were easily labelled due to tightly bound lexical fields, which enhanced the interpretability of overall results. The use of VADER for sentiment analysis proved well-suited for the informal, stylised nature of song lyrics. Its rule-based logic allowed effective processing of lyrics despite incomplete structure, idiomatic language, or stylistic punctuation. The method proved computationally efficient as it managed nearly 6,000 songs with fast processing speeds. Additionally, its dual output of compound sentiment scores and categorical labels enabled both quantitative and qualitative sentiment interpretations. The integration of sentiment across the full dataset enabled cross-temporal comparisons, broadening analytical possibilities without excessive reformatting. This approach could be further expanded by incorporating genre or artist information for deeper insights into musical trends. The hybrid approach of combining sentiment classification and topic modelling introduced a novel layer of insight into how emotional tone interacts with lyrical themes. By splitting the dataset into two sentiment subsets, clearer emotional distinctions emerged within each cluster after LDA. This approach helped isolate sentiment-driven themes that might have been blurred in general topic modelling, such as the contrast between empowerment positivity and introspective negativity. The resulting emotion-specific dictionaries highlighted not only thematic variation but also the depth of mood representation in popular music, making this method particularly effective for captur-

ing the layered nature of musical expression.

Limitations Despite the valuable insights gained, several limitations emerged in the application of these methods to lyrical data, highlighting the challenges of using text-based models to interpret an art form that is deeply contextual, metaphorical, and multimodal. While LDA was useful in identifying word-based groupings, it often failed to grasp the emotional tone or nuanced meanings within individual songs. Songs that shared vocabulary were not always thematically aligned, leading to misleading or incoherent groupings. This limitation exposes the challenges of relying solely on word frequency, particularly in a sensitive medium such as music. Moreover, genre, rhythm, and tone remain invisible to LDA, which reduces interpretation strength. Although VADER performed well in sentiment classification, it had limitations in capturing deeper emotional nuance. Song lyrics often use metaphor, irony, or layered poetic devices that a lexicon-based tool like VADDER cannot fully interpret. Cleaning procedures—such as stanza, marker, and punctuation removals—may have hindered the sensitivity to tonal shifts or intensity. Additionally, the frequent use of emotionally charged words can skew sentiment scores, which introduces the risk of oversimplifying sentiment interpretation where emotional complexity is often central. The hybrid approach aimed to address the general models’ shortcomings, but it also introduced challenges that must be acknowledged. Splitting the dataset reduced the volume of data for LDA, potentially weakening the coherence of topics. There is also a risk of overinterpreting sentiment-driven topics as they may appear distinct due to artificial separation rather than true emotional divergence. Furthermore, by excluding neutral sentiment from the analysis, potentially significant themes that straddle positive and negative expressions were omitted. This selective framing, while sharpening emotional focus, could limit broader representational power.

CONCLUSION

This study illustrates how computational techniques like topic modelling and sentiment analysis can uncover the emotional and thematic dimensions of popular music. By applying Latent Dirichlet Allocation and VADER sentiment analysis to lyrics, dominant themes and the nuanced emotional shifts in popular music over time were revealed. The hybrid approach, which separated sentiment before topic modelling, allowed for a richer understanding of how emotional tone shapes lyrical themes. Despite its strengths, the study highlights the limitations of relying solely on lexical data to interpret music’s deeply metaphorical and performance-driven context. Future work could explore genre-based segmentation, which may introduce distinct emotional layers and thematic conventions. Incorporating supervised learning may also improve accuracy and interpretation by training on emotionally labelled datasets. Overall, while computational tools offer unique insights, they require fine-tuning and grounding in emotional theory to meaningfully interpret lyrical data. Ultimately, this research opens pathways for more refined, context-aware analyses that better reflect the emotional complexity of modern music.

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APPENDIX

The complete python code for API usage, data collection, preprocessing and analysis is available at the following GitHub repository: https://github.com/rodriguesathena/QTA_Spring2025/tree/main/Final/api_based